MILITARY SPECIFICATION

LAUNDRY UNIT, TRAILER MOUNTED

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a trailer mounted laundry unit (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-C-375	- Circuit Breaker, Molded Case, Branch Circuit and Service
W-P-115	- Panel, Power Distribution
HH-I-553	- Insulation Tape, Electrical (Rubber, Natural and Synthetic)
NN-P-530	- Plywood, Flat Panel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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QQ-S-571 - Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy
QQ-S-781 - Strapping, Steel and Seals
MMM-A-260 - Adhesive, Water-resistant, (For Sealing Waterproofed Paper)
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-1055 - Barrier Material, Waterproofed, Flexible

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MIL-C-104 – Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 – Preservation, Methods of
MIL-B-121 – Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 – Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-B-844 – Blankets, Bed, Wool, Shrink Resistant and Mothproofed
MIL-B-3180 – Boiler and Related Equipment, Packaging of
MIL-C-3600 – Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of
MIL-V-13811 – Varnish, Waterproofing, Electrical, Ignition
MIL-T-22085 – Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing
MIL-H-44086 – Heater, Water, Liquid Fuel; M-80
MIL-D-44144 – Drying Tumbler, Laundry
MIL-E-44166 – Extractor, Laundry: Heavy Duty, Top Loading, Trailer Mounted
MIL-W-44167 – Washing Machine, Laundry, Open End Type, Trailer Mounted
MIL-G-52889 – Generator Set, Diesel Engine Driven 5KW, 60 Hertz and 10KW, 60 and 400 Hertz Tactical, Utility, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-195 - Marking of Connections for Electrical Assemblies
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1474 - Noise Limits for Army Material

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DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

6-1-9836 - Laundry Unit M 85
6-1-9374 - Finish Requirements CARC Paint

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA NO. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park Quincy, MA 02169.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

ICS1-110.13 Type 3R Enclosures - Rainproof and Sleet - (Ice) Resistant-Outdoor

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

UNDERWRITERS LABORATORIES, INC. (UL)

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMEICAN SOCIETY OF MECHANICAL ENGINEERS (AMSE)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF AMERICAN RAILROADS PUBLICATION

Pamphlet No. MD-7 - Rules Governing the Loading of Department of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, IL 60605.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Codes and standards. The laundry unit shall comply with the applicable requirements of NFPA 70 and NEMA ICS1-110.13 Type 3R.

3.3 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document (see 6.4).

3.4 Alternate components. Components offered as equivalent to components specified herein and on the reference drawings as a specific manufacturer's part number shall be functionally equal with the manufacturer's part number identified. The incorporation of such a component in the design of the specified end product shall not require modification to any other component, unless such modification is specifically approved by the contracting officer. Prior to commencing production, the contractor shall submit for the contracting officers' approval, a list identifying each alternate component, together with proof that

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each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.5 Design and construction. Design and construction of the trailer mounted laundry unit shall be as specified herein and on Drawing 6-1-9836 and all subsidiary drawings and parts list pertaining thereto.

3.5.1 Electrical requirements. The trailer mounted, laundry unit, electrical system shall be 60 cycle, alternating current. The power console shall be fed with a 60 amp, 3 phase, 208/120 volt power feed. The power will be distributed from a circuit breaker panel to fixed and remotely operated equipment. All electrical components shall be UL listed. All electrical work shall be done in accordance with NFPA No. 70.

3.5.2 Power distribution panel (see 6.5.3). A 3 phase, 4 wire circuit breaker panel board shall conform to type 1 of W-P-115. The enclosure for the panel shall conform to NEMA ICS1-110.13 Type 3R/12. The interior shall have 30 positions, a solid neutral and main lugs only. The breakers shall be the bolt on type conforming to W-C-375, sheets 10a, 10b, ha, 11b, 12a and 12b. The circuit breaker panel board and circuit breakers shall have a minimum integrated equipment rating of 22000 amps (root mean square value).

3.5.3 Wiring. Wiring specified herein and all wiring not delineated on the applicable detail drawings shall be done in accordance with NFPA Standard No. 70• All wires or wiring shall be identified in accordance with the applicable wiring diagrams and marked in accordance with MIL-STD-195. Unless otherwise specified on the detail drawings, all wiring connections shall be soldered, taped with rubber tape, or have staked connections. The solder shall conform to type R of QQ-S-571 and the tape shall conform to HH-I-553.

3.5.4 Moisture and fungus-resistant treatment. When specified (see 6.2), electrical components of equipment shall be moisture and fungus-resistant treated with materials conforming to and applied as required by MIL-T-152.

3.5.5 Drying, tumbler. The drying tumbler shall be in accordance with MIL-H-44144.

3.5.6 Heater, water. The water heater shall be in accordance with MIL-H-44086.

3.5.7 Washing machine. The washing machine shall be in accordance with MIL-W-44167.

3.5.8 Extractor. The extractor shall be in accordance with MIL-E-44166.

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3.6 Performance.

3.6.1 Electrical distribution system. Every end item shall be tested as specified in 4.6.1. Any test failure shall be cause for rejection of the end item.

3.6.2 Air distribution system. The air distribution system when operating for 15 minutes shall not fail and shall show no evidence of the following conditions when tested as specified in 4.6.2.

a. Overheated bearings.
b. Leaks from any part of the system.
c. Low pressure.
d. Inoperable valves.

3.6.3 Water distribution system. The water distribution system, the pump, water lines, and boiler when operating for 15 minutes shall not fail and show no evidence of the following conditions when tested as specified in 4.6.3.

a. Leaks from any part of the system.
b. Low pressure.
c. Improper operation of water heater and controls.

3.6.4 Washing machine. The washing machine and related equipment shall not fail or show evidence of the following conditions when tested as specified in 4.6.4.

a. Improper operation of controls.
b. Leaks during washing and rinse cycles.
c. Improper or incomplete operation of pressure controls, air valves, intake valves, or discharge valves.
d. Structural failures or failures of welds.
e. Improper water level.

3.6.5 Extractor machine. The extractor machine when operated shall not fail and shall show no evidence of the following conditions when tested as specified in 4.6.5.

a. Improper operation of controls.
b. Leaks from any portion of the curb or plumbing.
c. Structural failures.
d. Excessive vibration during operation.

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3.6.6 Dryer. The dryer when operated shall not fail or show any evidence of the following conditions when tested as specified in 4.6.6.

a. Improper rotation of drum, fan, or motor.
b. Leaks in fuel lines.
c. Structural failures or failures of welds.
d. Improper operation of controls.

3.6.7 Endurance test. The assembled laundry unit shall launder wool blankets when tested as specified in 4.6.7. There shall be no damages to any components throughout the test and all electrical and mechanical components shall function properly. There shall be no sign of warped or distorted metal and failed or cracked welds.

3.6.8 Road and terrain test. The assembled laundry shall traverse the test course, and show no damage to the components. The components shall remain securely fastened with no evidence of looseness, structural failure or operational failure, when tested as specified in 4.6.8.

3.6.9 Rail impact test. The assembled laundry unit shall be rail impact tested and show no damage to the components. The components shall remain securely fastened with no evidence of looseness, structural failure, or operational failure when tested as specified in 4.6.9.

3.6.10 Noise limits. The noise limits produced by the laundry unit shall meet the limits specified in MIL-STD-1474 and where applicable shall not exceed 85 decibels at the normal operators position when tested as specified in 4.6.10. If these limits are exceeded, noise level warning signs stating hearing protection requirements shall be permanently affixed to the laundry unit in proper locations (see Drawing 6-1-9836).

3.7 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.8 Government furnished property. The XM1061E1 cargo trailers, the 10 KW generators, water pumps, and the M80 water heaters will be Government furnished property and will be altered by the contractor if required, as shown in the drawing package (6-1-9836) see 6.2.

3.9 Welders. Before assigning any welder to work, the contractor shall provide the contracting officer, or his authorized representative, with the names of welders to be employed in the work, together with certification that each of these welders has passed a qualifications test as prescribed by either of the following listed codes for the type welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

a. Standard Qualification Procedure of the American Welding Society.
b. Welding Qualification of the American Society of Mechanical Engineers.

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3.10 Workmanship. The units shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malfunctions. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray sag, imprint or embedded foreign matter. Threaded fasteners shall not be broken, stripped or loose.

3.10.1 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds, and rough edges and projections shall be dressed smooth.

3.10.2 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide stain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and 4.4.3 and tested as specified in 4.6.1 thru 4.6.10. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish certificates of compliance for the requirements specified in 3.5.1 and 3.5.2.

4.4.1.2 In-process inspection. Examination of the surface preparation prior to welding shall be made to determine compliance with 3.10.1. Whenever nonconformance is noted correction shall be made to the applicable process and all affected items processed.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of laundry trailers. The sample unit shall be one laundry trailer. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified.		X
	Blistered, peeled, chipped, or area of no film.		X
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded.		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major		Minor
Design, construction and workmanship applicable to all components and assemblies	Any component or part missing or not specified type or size	X		
	Any component that does not function, malfunctions, or is misplaced to the degree that it does not serve its intended purpose	X		
	Not fabricated as specified by the document or drawing	X		
	Not clean (evidence of oil, grease, or dirt)			X
	Sharp edge or burr			X
	Puncture, fracture, or malformed	X		
	Design detail deviating from document or drawing requirements	X		
	Evidence of scale, rust, discoloration, or deposit of any foreign matter			X
	Finish omitted or not as specified:			
	a. On brass or aluminum components			X
	b. On steel components	X		
	Area of partial or no finish			X
Riveting	Mating surfaces of riveted components not drawn up tightly		*	
	Rivet head not in contact with its mating surface		*	
	Rivet loose - not properly set		*	
	Roll of a rivet contains more than one split			X
	Portion of a rivet is set or rolled on itself in lieu of being set or rolled on the mating surface of the component	X		
Welding	Missing, incomplete, cracked, fractured, or not fused	X		
	Undercut, not smooth and uniform, or scale or flux deposits not removed			X
Soldering (when applicable)	Missing, not adherent, or incomplete			X
	Not clean (flux residue not removed, pinholes in solder)			X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners	Broken, stripped, fractured, or not tight		X
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	X	
	Adequate slack not provided for wiring to relieve strain		X
Marking	Missing; incorrect; incomplete; illegible; of improper size, location, sequence or method of application	X	

NOTE: Defects designated by an asterisk (*) shall be scored as "major" when seriously affecting serviceability or appearance and "minor" when affecting serviceability or appearance but not seriously.

4.4.3 End item dimensional examination. The end item shall be examined for compliance with dimensions specified in this document and on the drawings. The lot size shall be expressed in units of laundry trailers. Any dimension that is not within the specified tolerance shall be classified as a defect. The sample unit shall be one laundry trailer unit. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units shall be 6.5.

4.4.4 End item testing. When a first article is not required, the initial production end item shall be tested as specified in 4.6.7. Each production end item shall be tested for the characteristics listed in table II. Failure of any test shall be cause for rejection of the end item involved.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Electrical distribution system	3.6.1	4.6.1
Air distribution system	3.6.2	4.6.2
Water distribution system	3.6.3	4.6.3
Washing machine	3.6.4	4.6.4
Extractor machine	3.6.5	4.6.5
Dryer	3.6.6	4.6.6
Endurance	3.6.7	4.6.7
Road and terrain	3.6.8	4.6.8
Rail impact	3.6.9	4.6.9
Noise limits	3.6.10	4.6.10

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4.4.5 Packaging examination. The fully packaged end item shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: loose strapping, components not secured, plywood cap not adequately bolted, bolts missing.
Preservation	Preservation improperly applied or missing.

4.5 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.4, shall be examined and validated as proof of compliance.

4.5.1 NFPA. Acceptable evidence of meeting the requirements of NFPA shall be a certificate of compliance certifying that the wiring of the laundry units conforms to NFPA 70.

4.6 Methods of inspection.

4.6.1 Electrical system test. To determine compliance with 3.5.1 thru 3.5.3, the electrical system of the laundry unit shall be checked, using the government furnished generator, to verify continuity of all circuits of the electrical installation to ascertain electrical readiness of the laundry. All electrically powered equipment shall be operated. Start all electrical equipment one item at a time and verify that the motor rotation is correct. If motor rotation is incorrect remedy the problem by switching any two of the phase conductors. Any evidence of incomplete circuits, non-operating switches or damaged electrical components shall constitute failure of this test.

4.6.2 Air distribution system test. Activate compressor and inspect air lines and fittings for leaks. Repair as required. Inspect pressure gage and verify that the reading is between 90 and 100 PSI. Individually activate each air valve from the washer control and visually verify that the valves mechanically function. Any nonconformance with the requirements of 3.6.2 shall constitute failure of this test.

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4.6.3 Water distribution system. Connect all hoses. Cap the end of the washer intake hose. Operate the pump drawing water from a nonpressurized tank. Any nonconformance with the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Washing machine test. With all the utility systems activated, manually operate the washer without clothes. Activate individually each intake and discharge valve. Drain the system between each intake test. Activate the remaining switches and verify that each operates. Upon completion of this, load the washer with 60 pounds of woolen blankets (type I or II of MIL-B-844) and launder using the automatic chart for wool. Any nonconformance with the requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Extractor test. Upon completion of 4.6.9, place the woolen blankets into two batches approximately 30 pounds each dry weight in the extractor. The extractor timer shall be set for three minutes and the extractor shall be operated through 2 complete cycles. Any nonconformance with the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Dryer test. Upon completion of the extractor test, place a 30 pound load in the dryer. With the drying temperature control set on 180°F, start the drying cycle and operate for 10 minutes. Any non-conformance with the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Endurance test. The laundry shall process woolens for 160 hours. This shall be accomplished in 8 consecutive 20 hour work days. The operation shall be at a rate of two completed laundry cycles per hour. The washer shall be operated in the automatic mode using the formula card for woolens. No detergent is required. Upon completion of the test, inspect all components including the washer, extractor, dryer, generator, water pump, water heater, air distribution system, electrical system and all supporting hardware. Any nonconformance with the requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Road and terrain test. The completely assembled laundry unit, shall be tested by the Government at a specified location (see 6.2). The test shall be as follows:

a. 125 miles over graveled roads at speeds averaging 35 mph. The graveled road test shall include not less than 20 quick starts and stops on grades of not less than 5 percent or greater than 10 percent. Fifty percent of the starts and stops shall be made on ascending grades and 50 percent on descending grades.

b. 125 miles over Belgian block or similar course averaging 10 mph.

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c. 125 miles over hilly cross-country terrain and 125 miles over level cross-country terrain at speeds averaging 10 mph. The open terrain and cross-country course shall have surface-irregularities consisting of mounds, rocks, brush and irregularities normally in cross-country or open terrain. The test drive shall include 20 abrupt starts, 20 sharp left hand turns, and 20 sharp right hand turns.

The laundry unit shall then be examined and the laundry shall be set up for operation and process woolens in 2 hours. The operation shall be at the rate of two completed laundry cycles per hour. Upon completion of the test inspect all components. Any nonconformance with the requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Rail impact test. The assembled laundry trailer unit shall be tested by the Government at a specified location (see 6.2). The test shall be in accordance with procedure IV of MIL-STD-810 and shall be as follows:

a. Forward impact at 8, 10 and 12 mph.
b. Rearward impact at 8, 10 and 12 mph.

The laundry unit shall then be examined and the laundry shall be set up for operation and process woolens in 2 hours. The operation shall be at the rate of two completed laundry cycles per hour. Upon completion of the test inspect all components. Any nonconformance with the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Noise limits test. The laundry unit shall be tested for conformance with the noise limits specified in 3.6.10 in accordance with MIL-STD-1474.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Preservation hereinafter specified shall be in accordance with the method and material requirements of MIL-P-116. Varnish hereinafter specified shall conform to MIL-V-13811. Tape hereinafter specified shall conform to MIL-T-22085. Surfaces to which tape is applied shall be free of dirt, grease, and any other foreign matter. The tape shall not be applied direct to plastic lenses, gauge windows, etc. Such surfaces shall be protected with kraft paper or fiberboard prior to application of the tape.

5.1.1.1 Drainage of water systems after factory testing. The entire water system of the units including the water heater, water pumps, water hoses, tumbler, washer extractor, etc., shall be thoroughly drained and parts blown out and dried with dry compressed air. All valves, petcocks, etc., shall be in the open position. A red tag shall be wired in a conspicuous place on the units indicating that the entire water system has been completely drained.

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5.1.1.2 Drainage of fuel system after factory testing. The fuel lines, water and tumbler heater, fuel pumps, fuel filter, engine carburetor, solenoid fuel valves, and all fuel connections shall be completely drained and thoroughly dried and replaced. The exterior surfaces of all flexible fuel lines shall be coated with varnish.

5.1.1.3 Cleaning. The complete laundry shall be cleaned process C-1 and thoroughly dried prior to application of preservative.

5.1.1.4 Painting. Painted surfaces on which the paint film has become damaged or defective shall be cleaned as specified in 5.1.1.3 and repainted using materials of the same quality and color as those removed.

5.1.1.5 Preservative application. All cleaned, uncoated ferrous metal surfaces shall be coated with a preservative. Surfaces that do not contact other surfaces in operation shall be coated with type P-1 preservative, and surfaces that contact other surfaces in operation shall be coated with type P-2, P-6 or P-7 preservative, except those components preserved as specified hereinafter.

5.1.1.5.1 Tumbler. All exposed, uncoated, ferrous metal surfaces of the tumbler, cleaned as specified, shall be coated with type P-6 preservative. The door switch handle and hinge shall be coated with type P-7 preservative. all exposed wiring shall be coated with varnish. The belts shall be removed, uncoated surfaces of the sheaves shall be prepared and finished per Drawing 6-1-9374, thoroughly dried and the belts replaced with tension relieved. The exhaust ports, control box and the burner air intake shall be sealed or covered with a combination of barrier material conforming to type I or II, grade A of MIL-B-121 and taped. All openings into and around switches, louvers, inlet and outlet fittings, access doors, receptacles, air vent, surfaces of door latch, timer, and thermometer face shall be sealed with tape. The timer shall be secured from movement with tape.

5.1.1.5.1.1 Caution tag. A red caution tag giving explicit reprocessing instructions shall be wired in a conspicuous place on the tumbler. These instructions shall include a warning to the effect that all tape and preservatives shall be removed and belt retensioned prior to putting the tumbler into operation.

5.1.1.5.2 Air receiver. The interior of the air receiver shall be drained of any accumulated moisture, thoroughly dried and fogged with type P-10 preservative. The drain outlet shall be closed.

5.1.1.5.3 Washer assembly. The cleaned and lubricated uncoated surfaces of nonmoving components of the washer shall be coated with type P-1 preservative. All inlets and outlets, shell door, soap trough cover, all valve handles in open position, timer, power takeoff, all switches in off position, safety switch cam

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arm, nameplate, temperature gauge face, etc., shall be secured and sealed with tape. Uncoated surfaces of sheaves shall be prepared and finished per Drawing 6-1-9374.

5.1.1.5.3.1 Extractor assembly. The cleaned and lubricated uncoated surfaces of nonmoving components of the extractor shall be coated with type P-1 preservative. All inlets and outlets, door, all valve handles in open position, timer, power takeoff, all switches in off positions, safety switch cam arm, nameplate, etc., shall be secured and sealed with tape.

5.1.1.5.3.2 Caution tag. A red caution tag giving explicit reprocessing instructions shall be wired in a conspicuous place on the unit. These instructions shall include a warning to the effect that all tape, preservative and coverings shall be removed prior to putting the unit into operation.

5.1.1.5.4 Fire extinguisher. All paint present on five extinguishers as procured must be stripped and all exposed ferrous surfaces shall be prepared and finished per Drawing 6-1-9374 except that the color shall be aircraft red No. 31136.

5.1.1.5.5 Washer control panel assembly. All wiring and uncoated surfaces in the control panel subject to corrosion, except electrical contact points, shall be coated with varnish. Varnish shall be applied in such a manner as not to affect the operation of the control panel. The formula control drums, shall be individually wrapped in barrier material specified in 5.1.1.5.1 and the warp secured with cotton tape or twine. All openings into the control panel assembly shall be sealed with tape.

5.1.1.5.6 Trailer chassis. Each trailer chassis shall be preserved in accordance with level A requirements of MIL-STD-281.

5.1.1.5.7 Engine-generator assembly. Each engine generator assembly, including batteries and electrolyte, shall be preserved in accordance with the level B preservation requirements of MIL-G-52889, except that the reference to securing the base of the container shall not apply.

5.1.1.5.8 Air compressor. The air compressor shall be preserved in accordance with the applicable requirements of MIL-C-3600.

5.1.1.5.9 Water heater. The water heater shall be preserved in accordance with the applicable requirements of MIL-B-3180.

5.1.1.6 Securing.

5.1.1.6.1 Water supply pump. The water supply pump shall be secured in place as shown on the applicable drawing.

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5.1.1.6.2 Electrical cables, hoses, suction strainer, support hardware and canvas cover. Detachable electrical cables and hoses shall be coiled and unit packed separately in snug-fitting fiberboard boxes conforming to type CF (variety optional) or SF, class domestic, grade 275, style RSC of PPP-B-636. The packed components shall then be placed as noted on the applicable drawing and secured in a manner as to prevent movement or damage while in transit. The canvas cover for the laundry unit shall completely cover the laundry assembly and be secured so as not to become loose during transportation.

5.1.1.6.3 One and one-half, five, and seven-inch exhaust ducts. The one and one-half inch exhaust ducts for the generator exhaust shall be placed within the 5-inch exhaust duct. The exhaust ducts shall be secured in a manner as to prevent movement during transit. The duct sections shall then be secured to the trailer chassis with the straps provided. The two 7-inch exhaust duct sections shall be placed in the overhead baskets and secured as shown on Drawing 6-1-9836.

5.1.1.6.4 Flexible 12-inch exhaust duct. The flexible 12-inch exhaust duct shall be completely collapsed and wrapped in 30-pound kraft paper or barrier material specified in 5.1.1.5.1 and the wrap secured with cotton cord, twine or tape. The wrapped duct shall be secured within the tumbler.

5.1.2 Commercial. The laundry shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. A close-fitting plywood cap shall be fitted over the laundry unit so that it rests on the trailer deck. The cap shall consist off two sides, two ends, and a top, and shall be constructed in accordance with type I, class 2 of MIL-C-104, except plywood shall conform to group A, type I, grade 3-4, or group B, with exterior glue, of NN-P-530 and lower frame members of the sides and ends shall be nominal 1 by 6 inch. Other frame members shall be those specified in MIL-C-104 for panel members for a net weight of 4000 pounds, except where interference with trailer components require that dimensions of members may be changed. The 1 by 6 inch lower frame members shall be laid on edge on the trailer deck and bridged to provide clearance for the trailer wheels and other projection on the trailer deck. The cap shall be positioned in such a manner as not to interfere with lifting eyes or damage the fabric cover. The cap shall be bolted to the trailer by running the plywood sheathing below the 1 by 6 inch bottom member, drilling holes through the plywood and trailer rail member, and bolting with 3/8 inch diameter zinc or cadmium plated machine bolts, flat and lock washers and nuts. Flat washers shall be located under the bolt head next to plywood, and lock washers shall be located under the nut next to trailer rail member. Four bolts shall be used on each side of the trailer, and two on the back. Plywood shall be cut out to provide clearance for reflecting lights and other protrusions on the trailer.

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5.2.1.1 Railway shipment. For railway shipment, tire pressure shall be increased 10 pounds over the required running pressure. Loading of vehicles for railway shipment shall be on open-top cars in accordance with Rules Governing the Loading of Department of Defense Materials on Open-Top Cars, Pamphlet MD-7 of the Association of American Railroads.

5.2.1.2 Highway haulaway shipment. The units shall be placed on bed or rails of haulaway vehicles so that the load is evenly distributed and secured. Parking brakes shall be locked and the levers wired or blocked in position. Loading of units for haulaway shall be in accordance with the Department of Transportation Motor Carrier Safety Regulations. Tire pressure shall be increased 10 pounds over the required running pressure.

5.2.1.3 Highway towaway shipment. No processing of the trailer units running gear shall be performed prior to highway shipment that would impair necessary operation or safety transit. Towaway rules for shipment by driveaway shall be in accordance with the Department of Transportation Motor Carrier Safety Regulations. Tire pressure shall be the required running pressure. Processing of the trailer units running gear, not accomplished prior to highway shipment, shall be performed at intermediate destination when vehicles are to be placed in storage or shipped overseas.

5.2.2 Level B packing. The laundry units, preserved as specified in 5.1, shall have all openings into electric motors that will permit entry of liquid water, dirt, or dust sealed with a contrasting color tape. All gauges, meters, instruction dials, sight glasses, and access door glass shall be covered with single-faced corrugated fiberboard or other cushioning material secured with commercial tape. The canvas cover for the laundry unit shall be placed in a heat sealed waterproof bag or wrapped in waterproof barrier material conforming to PPP-B-1055, prior to being unit packed in a fiberboard box specified in 5.1.1.6.2. All seams of the warp shall be sealed with a 3/4 inch minimum width continuous seam of water-resistance adhesive conforming to MMM-A-260. The canvas cover shall then be unit packed in a fiberboard box specified in 5.1.1.6.2, and secured to the laundry unit with at least two lengthwise and two girthwise straps, minimum 3/8 by 0.020 inch conforming to type I, class A or B of QQ-S-781.

5.2.3 Commercial packing. Laundries, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to the caution and reprocessing marking specified herein, interior packs and the laundry unit shall be marked in accordance with MIL-STD-129 or ASTM D 3951. Special requirements applicable to unpacked vehicles shall apply for level B and Commercial packs.

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6. NOTES

6.1 Intended use. The laundry unit is mobile for use in the field by personnel of the Department of Defense to provide laundry service to troop units and hospitals. The laundry unit is designed to take water from streams, rivers, lakes and the like by means of suction and to heat it sufficiently for laundering purposes. The laundry unit has a secondary function of decontaminating personal clothing from chemical warfare and bacteriological warfare agents and radioactive materials.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. When moisture and fungus-resistant treatment is required (see 3.5.4).
d. Source and delivery of Government furnished property (see 3.8).
e. Required location of 500 mile road test and rail impact test (see 4.6.8 and 4.6.9).
f. Selection for the applicable level of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.5 Sources of supply.

6.5.1 Washer. A washer conforming to Pellerir Milnor Corp. (P.O. Box 400, Dept. TR, Kenner, LA 70063) Model No. M 360260EW with Mitrol control unit, air actuated valves and water recycle controls and valves has been found to meet the requirements of 3.5.7 except the item must be modified to meet the dimensional, and other requirements of Drawing 6-1-9944.

6.5.2 Extractor. A extractor conforming to Bock Laundry Machine Co. (3600 Summit St, Station G, Box 5127, Toledo, OH 43611) Model No. 605 MIL has been found to meet the requirements of 3.5.8 except the item must be modified to meet the dimensional, and other requirements of Drawing 6-1-9945.

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6.5.3 Power distribution panel. A power distribution panel conforming to Square D-Company (Palatine, IL) Interior Model No. NQOB-30429-1 with enclosure No. MH 29WP, interior trim kit No. NQ 29 MSK, panel board interior mounting bracket No. NQM-25D, 20 AMP breaker No. QOB 320 and 60 AMP breaker No. QOB 360 has been found to meet the requirements of 3.5.2.

6.6 Subject term (key word) listing.

Cleaning, clothes
Laundry

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 3510-A296

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-44142A(GL)	Laundry Unit, Trailer Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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